



Special Requirements of Electronic Health Record Systems in Pediatrics: Clinical Report

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Pediatricians' use of electronic health record (EHR) systems has become nearly ubiquitous in the United States, yet many systems lack full functionality to deliver effective and efficient pediatric care. This clinical report seeks to provide a compendium of core pediatric functionality of importance to child health care providers that may serve as the focus for EHR developers and clinicians as they evaluate their EHR needs. Also reviewed are important but less critical functions, any of which might be of importance in a specific pediatric context. The major areas described here are immunization management, growth and development, social drivers of health tracking, decision support for orders, patient identification, data normalization, privacy, and system functionality standards in pediatric contexts.

INTRODUCTION

The technological landscape, including data sources both intra- and extramural to health care, continues to change rapidly, introducing advancements and challenges.^{1–7} Although pediatrician use of electronic health record (EHR) systems has become nearly ubiquitous in the United States, with nearly 94% of pediatric offices surveyed reporting EHR use in 2016, fewer than 17% reported using EHRs with full pediatric functionality at that time, which would include basic functionality, like weight-based dosing, and full functionality, like tracking immunization schedule adherence.⁷ Core pediatric-specific functionalities informed the 21st Century Cures Act in 2016, which established a voluntary certification of health information technology for pediatric care.^{5,8}

This clinical report seeks to provide a compendium of the specific functionalities that facilitate providing effective and efficient pediatric care. This report does not seek to enumerate administrative functions

abstract

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in the practice-management system (such as appointment management, insurance eligibility determination, and billing).

Information management is a key requirement for an effective medical home, which was envisioned by the American Academy of Pediatrics (AAP) in 1965 but required EHRs to become a reality.^{5,9,10} The Patient-Centered Medical Home Recognition program of the National Committee for Quality Assurance demands that physician practices manage information for the benefit of patients over the continuum of care. Although the National Committee for Quality Assurance does not require the use of an electronic system, the committee encourages the use of certified EHR products, recognizing that the functions required for Patient-Centered Medical Home certification are difficult to achieve without a sophisticated EHR.¹¹

PEDIATRIC FUNCTIONS

In 2001, the AAP published a description of the features that would be desirable in a clinical information system to be used in pediatrics.¹² Almost none of these features were purely pediatric but could provide value to other specialties as well (eg, geriatrics). Certainly, medication dosing by weight and recording guardianship information have examples in both pediatric and adult care; however, because these features are vastly more prevalent and highlighted in pediatrics, we refer to them here as “pediatric functions.” Although an ideal EHR would include at least the functions mentioned in this report, we have attempted to use an ordinal system to grade the importance of these functions from “may” to “should” to “must.”

CRITICAL PEDIATRIC EHR FUNCTIONAL AREAS

Several functional areas are so critical to the care of infants, children, and adolescents that their absence would impede quality pediatric care. These include the major areas described here, including immunization management; growth, development, and social drivers of health tracking; decision support for orders; patient identification; data normalization; privacy; and system functionality standards in pediatric contexts.

RELATION TO 21ST CENTURY CURES ACT

Several studies reviewing pediatric features in outpatient EHR systems demonstrated a significant disconnect between the features highlighted by the AAP as required for pediatric care and features that were used in pediatric offices throughout the country.^{7,13} In no small part because of efforts by the AAP, in 2016 the United States Congress passed the 21st Century Cures Act, which included the requirement for the Office of the National Coordinator to establish a voluntary pediatric certification of EHRs.⁸ Subsequently, AAP experts worked with the

Office of the National Coordinator to establish criteria for this certification. Once the Drummond Group began developing the certification, AAP experts participated in the program’s design.¹⁴ Although the certification will not test for all functionalities the AAP supports in the safe and effective care of children, it will be an important step into improving pediatric functionalities in EHRs.

IMMUNIZATION MANAGEMENT

EHR software used in pediatrics must be able to manage the dynamic complexities involved in the scheduling, recording, analyzing, and reporting details accurately and efficiently for a myriad of childhood immunizations. Being a dynamic and nimble system is critical, as the EHR must adjust its decision support and reporting details based on current recommendations, time of year, and a practice’s purchase and supply of immunizations.

The EHR can also help to document immunization irregularities. The EHR must contain the details needed to report any adverse events to the Vaccine Adverse Event Reporting System, and such events should remain a part of the patient’s immunization record.^{15–17} Documentation of purposefully skipped vaccines because of clinical contraindication or refusal by a caregiver or patient should also remain with the immunization record, along with an explanation of why the vaccine was missed.¹⁸ Including these data in the immunization record will allow for a single source of documentation for both received and not received vaccines, data that may be used to inform subsequent immunizations and care.

Linking to Immunization Information Systems

Immunization information system (IIS) technology has developed significantly in the last decade. Meaningful use incentives for immunization interoperability helped to drive this development, and the Centers for Disease Control and Prevention (CDC)’s National Center for Immunization and Respiratory Diseases helped by providing guidance through the Immunization Information Systems Strategic Plans.¹⁹ Although the focus of an IIS is interoperability of immunization systems, the CDC provides many details that may be helpful for EHR developers and consumers for local immunization management and for integration with remote systems, along with details and standards found in relevant AAP policy statements and technical reports.^{20,21}

Most states and several local jurisdictions have electronic IISs or registries.^{22–24} The EHR should allow interoperability with these systems, including the ability to download, upload, and synchronize a child’s immunization history, including from multiple disparate sources.

Immunization Decision Support

Systems for encoding rules about which immunizations are due and when they are projected to be due in the future have existed for years.²⁵ For an EHR system to fully support pediatric practice, it must be able to take previous immunization data and the patient's medical history and derive, at the point of care, logical conclusions about the current immunization status, recommend appropriate immunizations, and warn against any contraindications. This forecasting requires an understanding of the individual antigens present in each vaccine and analysis of when, in what form, and at what age in the child's life each antigen was—or was supposed to be—administered to ensure historical doses are valid. There may also be variations in this functionality based on individual patient health needs, such as adjustments for preterm birth, prior immunization reactions, immunoglobulin administration, or immunodeficiency. These functions might be built into the system or be derived from immunization registries or third-party programs accessed via a network. If the logic is built into the EHR system itself, there should be an easy way to update the logic to reflect changes to immunization rules or local epidemiology and to handle new vaccines and new antigen combinations. If the EHR consistently uses an IIS's forecasting tools, this forecasting might take precedence over EHR-derived forecasting.

Population Level Immunization Management

An EHR supporting pediatric care should facilitate reporting of pediatric-specific quality measures, such as immunization management across a population, such as a practice. Generating reports and snapshots based on immunization status may encourage proactive recall and engagement to close immunization care gaps. This may also help to inform proactive ordering and inventory management to ensure no opportunity is lost to vaccinate at the point of care because of inadequate stock.

GROWTH, DEVELOPMENT, AND SOCIAL DRIVERS OF HEALTH TRACKING

Central to the role of pediatric clinicians is that they are caring for patients in a dynamic state of growth, development, and social situations. Unlike adult care, the most important information is not the measurement itself (eg, height, weight, development) but its rate of acceleration, which must be compared with age-appropriate norms (eg, by recording and presenting a percentile or z-score) to provide meaningful information to pediatricians. The EHR must support longitudinal management and visualization of these data.

Graphical Representation of Growth

Clinicians make important judgments about a child's health by visual inspection of growth charts, plots of a child's body measurements (usually a combination of weight, length or height, head circumference, and BMI) over time. An individual's measurements are usually plotted against standard percentile curves, such as those from the CDC or World Health Organization.²⁶ These plots must be readily available to clinicians during the workflow of a standard patient visit. Percentile calculations and decision support based on historical values should be available to detect errors during value entry. The EHR system should allow the representation of percentile curves from a usual source or others that may provide adjustment for prematurity or other specific populations.^{27–29} The EHR should allow magnification (“zooming”) of the plot to inspect specific areas, differentiating height from length, derivation of growth-velocity values between data points, and display of percentile or z-score of a value in comparison with the reference standard selected. Ideally, mechanisms would exist to tag specific values as spurious and to filter based on criteria like location the measurement was obtained, such as might be desired in a pediatric nutrition clinic that seeks to obtain very accurate and precise weight, length, height, and head circumference data.³⁰ These data and visualizations should be available for caregivers through a patient portal.

Developmental Tracking

Pediatric clinicians use a variety of developmental screening tools throughout a child's life, including both surveys and direct assessment.^{31–33} Identification of children with developmental disabilities is necessary to implement early intervention services. The EHR should support patient- and provider-facing structured data acquisition from provider-supplied forms and screening tools, including a means of tracking use of those that require payment for licensed use. Although the EHR developer itself is not expected to supply these forms, a method should exist to import them from an external data source or, at a minimum, allow documentation of the aggregated scores and required follow-up actions. In some cases, audiovisual recordings may help augment developmental assessment documentation, such as those used for recording general movement assessments in preterm infants.³⁴

Collecting some developmental, social, and other information in a previsit survey has now become commonplace. This survey can come from the EHR's patient portal or through third-party software. Regardless of source, this information must be captured and imported into the EHR in a structured manner that supports review of trends over time.³⁵ Capture and use of these data must support workflows for caring for the individual patient during a clinic visit, chart review, and billing and coding, and also reduce administrative burdens for population management.

Social Drivers of Health

Since the early 2000s, the AAP and other groups have advocated for standardized screening for social drivers of health.^{36–38} The US Department of Health and Human Services Office of Disease Prevention and Health Promotion groups social determinants of health into 5 domains: economic stability, education access and quality, health care access and quality, neighborhood and built environment, and social and community context.³⁹ Because of the important role these factors play on a child's health, well-being, and quality of life, the EHR should support mechanisms to document, visualize, and provide decision support based on these factors throughout a child's life. Accurate capturing of race and ethnicity data can establish baseline data to address existing health disparities in the practice of pediatrics, with special attention to privacy and confidentiality of such data. EHR integration of social drivers of health that addresses parents' and children or adolescents' information is key to not only understanding barriers to achieving health equity but also to connect patients and families to available resources and document interventions taken by providers to eliminate health disparities. Because of the sensitive nature of these data, special attention must be given to privacy and confidentiality of the data.⁴⁰

FACILITATE DECISION SUPPORT FOR ORDERS

One key argument for the benefits of using an EHR is that it can improve safety of medical orders.^{2,41} Although most EHR developers do not wish to develop the rules contained in order entry decision support, they should support a mechanism that allows for decision support recommendation integration from multiple sources into a clinician's workflow for inpatient as well as outpatient orders. These sources could be rules imported directly into the EHR or clinical decision support service protocols, like CDS Connect and CDS Hooks.^{21,42,43}

MEDICATION DOSING

For medications, the EHR must provide a mechanism to integrate information from pharmacological databases that provide recommended biometrically appropriate and technically deliverable doses.⁴⁴ Suggested dosing parameters, rounding factors, calculation steps, biometric data used (eg, weight, body surface area, gestational age, or chronologic age), and any special comments or considerations should be displayed clearly through the EHR interface and persist with the order, whether transmitted via paper or electronically. Systems should have checks to ensure that the biometric data are current relative to expected changes for the child. For the case in which dosing guidelines or formulary benefits vary with age or gestational age, the system should incorporate those data into its decision

support.⁴⁵ If a historical weight is used to estimate a current weight for prescriptions, the EHR should support methods for the clinician to determine the validity of the estimate, understanding that an infant's weight will generally fluctuate more rapidly than an adolescent's.

Pediatricians must often write prescriptions in which the medication is divided in two or more labeled packages—one for home administration and one for administration during the day at school, child care, or another care setting. EHR systems should provide the capability to generate instructions to the pharmacy to dispense medications in this way, along with administration instructions to teachers and child care workers.

PATIENT IDENTIFICATION

Children undergo name changes because of the need to change from a temporary name assigned at the birth hospital, changes in family structure, or changes in preferred name. Because clinical data may be connected to old names, EHR systems should support retrieval of data via search on previous names. Ideally, these prior identifiers would transfer across disparate health systems.

Although some EHR systems depend on the use of a government-issued identification number (usually the Social Security number), newborn infants do not receive these numbers for a while after birth. EHR systems must allow the registration of patients without such identifiers and allow retrieval of information based on any temporary identifiers that may be used.

An EHR system that allows storage of prenatal data (eg, from a fetal imaging or surgical procedure) should allow the logical connection of these data to the postnatal record once the child's record is established in the system.

Ambiguous Sex

In the case of a child with ambiguous genitalia, an EHR system should allow the assignment of sex as unknown and to operate normally until the sex of the patient is assigned. Any decision support that relies on patient sex should allow operation with this ambiguous assignment.

Gender Identity

Similar complexities exist in the assignment of sex under conditions of varying gender identity. Systems should be able to store sex assigned at birth, current gender identity, and data on applicable anatomic and physiologic characteristics so that decision support based on sex continues to apply accurately. For example, the requirement for a negative pregnancy test before a procedure should apply if the patient retains a uterus. Clinicians should be able to select and use norms for growth and laboratory values based on applicable physiology. Information surrounding gender may also be sensitive information that requires special provisions to support privacy.

NORMS FOR PEDIATRIC DATA

Numerical Data

Normal values for almost all numerical data (eg, laboratory results, body measurements, vital signs, and scores on standardized assessments) change continuously with age and growth, so it is insufficient to provide merely a handful of normative ranges. Visualizing data by plotting values over time will often facilitate data analysis and decision-making, such as for laboratory values like bilirubin in a newborn infant. Developers should assume that all numeric data collected in a pediatric context have changing norm values over the lifespan and should provide ways of flagging abnormal values and inappropriate acceleration or deceleration of consecutive values at any age. Percentile values and z-scores (number of SD from the mean) should be available for those data for which the distributions are known, such as height, weight, head circumference, and BMI.

Data Precision

To support pediatric practice, an EHR must be able to handle data at appropriate numerical precision and graphical resolution, which may change over time. Body weight to the nearest gram is commonly accepted as an appropriate precision in neonatal facilities, whereas a tenth of a kilogram may be more appropriate for older children. As another example, an EHR system may present growth curves of height, weight, and head circumference, complete with appropriate normative curves for comparison. However, if those curves are available in only one graphical resolution, measurements obtained frequently (daily weight measurements, weekly head circumference measurements, etc) may become impossible to analyze visually. Age in the newborn nursery should be available in units at least down to the hour, if not to the minute. The units for age (hours, days, weeks, months, years) need to grow with the age of the child, as appropriate. Developers of EHR systems should consider how the small changes in numeric data that one sees in the care of young patients affect data recording and display.

Nonnumeric Data

Whenever an EHR system distinguishes normal from abnormal in nonnumeric data (eg, flagging the presence of a physical sign as abnormal), it should consider age in the interpretation of normality. For example, a system that considers “unable to feed self” a universally abnormal finding in the interpretation of a functional assessment is not taking the developmentally normal capabilities of young children into account. Another example is “fall risk,” which should be able to transition dynamically from a “drop risk,” to “developmentally appropriate falls,” and eventually to a true “fall risk.”

Complex Normative Relationships

Not all normative data are based solely on age. In the cases of blood pressure, intracardiac measurements, or peak flowmeter norms, values may be determined by age, sex, and height percentile.^{46,47} Therefore, clinical decision support that flags abnormal values based solely on chronologic age are insufficient for blood pressure and peak expiratory flow and may be insufficient for other measurements in pediatric patients. An appropriate flag will include the decision logic and reference the source for the concern.

Gestational Age

For neonates, chronologic age (expressed as the time since birth) is insufficient for medication-prescribing decision support, normative ranges for laboratory data, normative definitions for physical examination findings, vital sign ranges, and guideline-application support. Gestational age, chronologic age, and corrected age are each unique and important ways to present the age of a neonate⁴⁵; EHR systems should support each of these concepts for age and allow for their use in decision support.

PRIVACY

Supporting privacy for children and caregivers requires granular metadata and segmentation built into every facet of the EHR, as enumerating every potential privacy concern is impossible.⁴⁸ Even areas as seemingly innocuous as body weight or immunization status could lead to unintended disclosure, such as in a patient with an eating disorder or when hepatitis B or human papillomavirus vaccinations are administered after sexual exposure. Implementation considerations must ensure data retain their granular security and encryption wherever stored or transmitted. For example, medication, implanted device, and problem lists should retain their security and segmentation even when pulled into a clinical note so that these data are available to the medical providers who need them for directing care but can be kept private as necessary. Supporting child privacy will require special attention to differential access to medical data, especially with regard to adolescent and maternal data, family and social history, cases of abuse, and other child-specific situations described below.

Adolescent Privacy

Laws about age of consent vary from state to state and depend on the presenting problem.^{49–51} Adolescents seeking treatment of mental health disorders, for example, may consent to their treatment at an earlier age than the age of majority in most states.⁵² Some states also have laws regarding parental notification whereby interpretation is based on the patient's age and presenting problem.⁵³ Practices that serve adolescents typically have policies with respect to what portion of an adolescent's care should be handled with special

privacy protections (eg, in some jurisdictions, the adolescent must give explicit permission for the parent to review their records). These privacy protections may require the flagging of protected information; therefore, EHR systems should support data segmentation and privacy policies that vary by age, according to presenting problems, diagnoses, and orders (ie, medications, laboratory tests, and imaging studies) and be flexible enough to handle the policies of individual practices.

If an EHR system handles record keeping for consent for treatment, it should provide for the recording of assent for treatment (from an underaged adolescent or child) combined with parental informed consent^{54,55} as well as consent for treatment (from an adolescent) combined with a record of parental involvement.⁵⁴ The separation of the patient's consent or assent and the parent's or guardian's consent is particularly important in the areas of testing for drugs or alcohol, screening for sexually transmitted infections, outpatient mental health care, and prenatal care and delivery services. These circumstances may also be impacted by local jurisdiction, which may additionally draw a distinction between the right to consent for services and the right to privacy for health information related to those services.^{56–58}

Enabling appropriate external access to adolescent health information continues to be one of the largest challenges pediatricians face in offering patient portal access to their patients, especially with the stringent information-blocking restrictions imposed by the 21st Century Cures Act.^{8,48,59,60} Risks of privacy violations need to be transparent to the adolescent patients, and any disclosure should occur with the assent of the adolescent. EHR and portal default privacy settings should be strict, comply with state and federal laws, and facilitate privacy throughout the health care information journey.^{61–64} Unfortunately, sensitive information is often transmitted in ways that fail to maintain its sensitive status (eg, when a report is scanned as an image or when revealing billing information, such as sexually transmitted infection or pregnancy testing, is sent to a parent or guarantor).

Unfortunately, research has shown that more than half of adolescent portals were likely accessed by the guardian using the adolescent's credentials, diminishing the results of segmentation efforts.^{65,66} Authentication to ensure that the individual logging in is not an imposter—even a parent—is important.⁶⁷ An EHR system that shares potentially sensitive data must use modern identity confirmation techniques (eg, 2-factor authentication), preferably with biometrics in combination with questions that can only be answered by the correct user, in addition to the common practice of usernames and passwords. Although the easiest workaround is simply to disallow external access to adolescent health information, this approach prevents adolescents from communicating in one of their preferred methods. Interacting with adolescents

via a portal may empower them to take ownership of their own health and aid transition to adulthood.⁵⁹

Adolescent medical records will contain some information that must be shared with caregivers to allow them to care for the adolescent's medical needs, such as evaluation and treatment of most acute and chronic conditions. The EHR portal should allow differential (segmented) access protection to various data. Caregivers and adolescents should each have discrete login credentials, which would in turn provide tiered data access.^{68,69} Ideally, the adolescent should be able to control the information dissemination directly within the portal or its mobile application.

Maternal Data in the Child's Chart

The AAP has great concerns that sensitive maternal information (eg, opioid use, domestic violence) may be disclosed inappropriately, such as to another caregiver through a patient portal.^{8,70} Allowing access to this information may violate the parent's privacy and put both parent and infant at risk, which would reduce the likelihood of seeking medical care in the future. At a minimum, maternal data and potentially inferential data, like neonatal opiate withdrawal scores, neonatal testing, or treatment of sexually transmitted infections, should retain metadata tagging and encryption that can trigger additional safeguards surrounding disclosure. Although defining these safeguards lies beyond the scope of this report, ensuring such tagging exists and is appropriately retained during information transfer is critical to allow safeguards to be built, including those that may eventually limit a child's access to nonessential maternal information once they are able to access their own records.

DYNAMIC CAREGIVER STRUCTURES

Children can have a variety of caregivers, including parents, grandparents, step-parents, foster families, staff at group homes, and a host of other possibilities. Any of these individuals or other associated adults may accompany a child for a health care visit. The variety of caregivers generates uncertainties about who may provide and receive medical information and who may consent for medical treatments and immunizations. The EHR must support the ongoing documentation of a wide complex of dynamic family and living structures and must clearly indicate who is allowed to receive medical information and who can make decisions for the child, and it must keep a historical record of these details as living circumstances change.

Children in Child Welfare Protective Custody

When a child is removed from their family of origin and placed in child welfare protective custody such as foster care, complex issues of consent and confidentiality of medical information arise.^{71,72} The child will be living with adults who do not hold custody of the child and will

often present for medical care with individuals who do not have the ability to provide medical consent or receive all medical information. The EHR should support documentation of medical decision-making and health information sharing abilities and limitations for those involved in the child's care, including who has authority to consent to treatment, with references to appropriate legal documentation as necessary. An EHR system should allow reasonable access to custodians and court appointed attorneys to the medical information within the confines of applicable state law and privacy regulations. Although caregivers may not be granted access to the complete medical record, they ought to be able to communicate about appointments and access the minimal amount of information necessary to manage the health care of foster children in their care (eg, accessing immunization records in preparation for school enrollment and being aware of prescriptions or other medical treatments). Such access will require careful implementation of patient portals specifically designed for use by caregivers, with appropriate limitations on the duration of access and sufficient authentication methods to prevent inappropriate access. When a child is in the temporary custody of child welfare, the family of origin typically maintains rights to receive medical information and may need to provide consent for certain medical decisions (eg, surgeries). Family of origin must be considered in the careful implementation of information sharing in patient portals, supporting bidirectional confidentiality for family of origin, caregivers, and placement location. Access must remain dynamic to allow changes to access if a safety concern arises or the child's legal status changes, such as restoring full access to a family of origin when a child is reunited and custody is returned or terminating access if parental rights are terminated.

Adoption

Records of children undergoing adoption proceedings and those who have been adopted may need special privacy handling. In cases where state law offers special protections for the identity of birth parents, EHR systems should allow flagging and special privacy protection. In accordance with state laws, preadoption records may need to have names or other identifying information redacted or permanently locked. In situations in which an adopted child is not initially made aware of their adoption, it is important to recognize that the information ultimately belongs to that adopted person when they reach the age of majority.

Consent by Proxy

Children often present for nonurgent health care in the company of an adult who is not the custodial parent or guardian. The best way to prevent confusion about consent for care in this situation is to record the custodial parents' wishes as to which adult can consent to which

child's care and under what limitations.⁷³ EHR systems that manage consent for treatment should support recording this information.

EMERGENCY TREATMENT

When EHR systems support the recording of consent and assent for treatment, they should be flexible enough to allow for the emergency treatment of minors, in which the parent or legal guardian may be absent, and the usual procedures for consent must change.⁴⁹

OTHER PROTECTED POPULATIONS

The EHR should support special privacy protection for cases of child abuse and neglect. The Health Insurance Portability and Accountability Act (45 CFR §164.502(g)) allows for preventing access to health information by parents or guardians when the child or adolescent has been or may be subject to abuse or neglect by that person or if disclosure of this information is not in the best interest of the child or may endanger them. The EHR should have systems in place to prevent access to protected images taken during a medical evaluation for potential physical abuse, sexual abuse, or sexual assault.

GENETIC INFORMATION

Genetic data, because of the potential for stigma, discrimination, or bias if public, are challenging for providers. Genetic analysis can be beneficial for patients because it may uncover diagnoses otherwise missed and inform subsequent treatments. However, broad genetic testing might also identify genes that are clinically important but not actionable in childhood, such as a BRCA mutation. Many genetic tests of a predictive nature (eg, Huntington disease or Lynch syndrome), once disclosed, have significant repercussions to the life of the individual and their family. Genetic testing may also reveal nonpaternity, leading to significant effects on family dynamics.⁷⁴ Although research continues in this specific area, at a minimum, the EHR should facilitate an embargo on genetic information.

ACCESS TO EXTERNAL INFORMATION

One of the key requirements for an effective EHR is the ability to interact with and digest external data, including both from other clinical sources (eg, a hospital or clinic) and from device- and patient-generated data. To operate effectively as a medical home for children, primary care providers must be able to access subspecialty and care support encounter notes. Pediatricians may find particular benefit in systems that can connect with nonhealth system sources, such as schools, community health partners, and others.

Another form of interoperability would allow patients the opportunity to upload personal data from fitness trackers, weight scales, blood pressure monitors, glucometers, and other health-connected technology. Pediatricians may want to consider implementing systems that have a framework for accepting these data. For medically complex children with technology dependency, the integration of data from their devices is critical. The care team should be able to see their data in flowsheets and align them temporally with other time series data to ascertain a complete view of the patient's health.

Those wishing to design software available for children must follow relevant guidelines from the Children's Online Privacy Protection Rule.⁷⁵ These services should also function without requiring the child to have their own phone. Apps that integrate must be able to recognize proxy integration. If the target is the parent but needs the child's information, the app must be careful as to which is used. If left unchecked, an app that interprets data outside an appropriate pediatric context may lead to patient harm. Caregivers must understand both the security and privacy risks of allowing access to a minor's data. EHRs can help by providing plain-language guidance about potential risks.

PEDIATRIC EHR SYSTEM FUNCTIONALITY STANDARDS

Health Level Seven International, a not-for-profit health care data-standards organization, created the Health Level Seven International EHR-System Functional Model specification to define the functions for EHR system functions through its Electronic Health Record Technical Committee.^{53,76} This functional model was used as the basis for the Child Health Profile, which attempted to describe the modifications to the model needed to create an effective tool for child health.⁷⁷ This standard is being used as the basis for the EHR system certification process specified by the federal Office of the National Coordinator for Health Information Technology. The purpose of certification is to set a minimum level of functionality that EHR systems will have to meet to qualify for special treatment, such as participation in pay-for-performance programs.^{8,51,52}

EXTENSIBILITY

Although electronic health record usability has improved throughout the past decade, there remain several avenues for innovation, particularly in pediatrics. One specific opportunity would be to create a standard framework to integrate clinical practice guidelines into electronic health records. This would support efficient implementation of clinical practice guidelines in the pediatric practice, which can significantly enhance the quality of care of patients.

The Substitutable Medical Applications, Reusable Technologies initiative is a specific example of a framework

that an EHR developer can use to support substitutable applications or web services embedded in the EHR.^{21,78} Some of the earliest functioning applications that use this paradigm are pediatric focused, often building functionality for areas where standard EHR implementations have fallen short, including an interactive growth chart app, an interactive neonatal hyperbilirubinemia decision support tool, and a child blood pressure percentiles application.⁷⁹ The substitutable nature of these applications eliminate the need for pediatricians to develop the content and can empower individual pediatricians and groups to use quality resources developed by others and find the best fit for their practices.

ARTIFICIAL INTELLIGENCE AND PRECISION MEDICINE

Machine learning and artificial intelligence represent a variety of algorithms in which data are used to train computer algorithms to solve complex tasks. Machine learning can be used to process heterogeneous data types including numerical data; imaging data such as 2-dimensional, 3-dimensional, and video data; written text; spoken text; and streaming data. These algorithms can be very powerful, but their efficacy and best practices are currently being assessed. These tools may or may not have validated pediatric approvals and should be managed with caution.

Transfer learning is a technique that uses results of existing models to bootstrap learning for new tasks. A model learning to distinguish pictures of airplanes from birds might benefit by pretraining with results from one trained to distinguish automobiles from pedestrians. A danger inherent in transfer learning and machine learning in general is that it can introduce inductive bias into a system.^{80,81} A model that performs well in adult medicine might generate surprising results in a pediatric population. Any artificial intelligence or machine learning model being considered for implementation in pediatrics should be trained and benchmarked with representative pediatric data to help mitigate any biases inherent in the model. However, such models and applications must also consider special privacy protections surrounding the storage and use of a minor's data, such as might be generated while using a digital scribe.

TELEHEALTH CAPABILITIES EMBEDDED INTO EHRs

EHR systems are no longer only documentation, data repository, and ordering systems. They have become "holistic" in the sense of promoting a better and more efficient encounter experience between the patient and care team. To that end, some EHRs have added telemedicine capabilities. Pediatrics presents some unique care scenarios that use telemedicine. For example, school-based health care can benefit from telemedicine capabilities that allow subspecialists and primary care physicians to communicate with school nurses. Expanding telehealth capabilities in the EHR may also help provide subspecialty pediatric care to patients

and geographic areas that have traditionally been difficult to serve.

OTHER PEDIATRIC FUNCTIONS

This report outlines the major areas of functionality that are relatively more important in pediatric care than in adult care. There are, of course, many other functions that are important, including the following:

- Archive and manage patient data for a statutorily defined period.
- Provide educational materials that are appropriate to parents and children and at varying reading levels.
- Create pedigree diagrams.
- Display age and weight at all times throughout the user interface.
- Select age-based documentation templates and order sets based on a patient’s age.
- Indicate whether a guideline applies to a patient based on age.
- Indicate the source of patient data, especially when the source is not the patient or the parent (eg, the schoolteacher or child care worker).

THE FUTURE OF THE PEDIATRIC EHR SYSTEM

In the wake of the rapid uptake of EHR systems in the years since the first AAP statement,^{7,13,82} national groups have expressed increased interest in standardizing the features of EHR systems and certifying their functions.⁸² Child health care providers want to ensure that pediatric functions, terminology, and data precision are built into these standards and certification processes. Pediatric care providers want this not only to make their own systems more effective in improving the health of children but also to make all EHR systems more useful for patients of all ages.

The AAP is working proactively to ensure that knowledgeable pediatricians who can thoroughly explain child health care issues are invited to address the groups that set these standards. This report should serve as a guide for these efforts to represent the interests of child health care providers and present a guide to individual practitioners who are evaluating a given system’s ability to perform in the pediatric environment.

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ABBREVIATIONS

- AAP: American Academy of Pediatrics
- CDC: Centers for Disease Control and Prevention
- EHR: electronic health record
- IIS: immunization information system

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